

ONE WALL IN MANY COORDINATES: CLOSURE OF THE PROPAGATION TRICHOTOMY AND THE MASTER QUANTIFIER FOR THE RIEMANN HYPOTHESIS

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ABSTRACT. The companion papers [6, 7] reduced the search for a proof of the Riemann Hypothesis (RH), within the Connes–Consani–Moscovici framework, to a trichotomy of argument-forms surviving a structural no-go: two-dimensional propagation (Form A), index amplification (Form B), and an effective self-approximation criterion (Form C). All three are now closed. We collect the three closure theorems and make the paper’s central claim: they are not three independent failures but one obstruction expressed in three coordinate systems. In each case the unconditionally controlled object is an *averaged*, *generic*, or *interior* quantity, and RH is equivalent to an *individual*, *uniform*, or *boundary* refinement of it; the wall is always the inversion of a single quantifier, from “on average / for almost all / in the interior” to “for the specific / uniformly / at the boundary.” We show this quantifier inversion is the same object as the parity barrier that closed the ω -class arc, the arithmetic-propagation wall MW-2, the Hadamard barrier MW-7, and the individual-versus-generic orbit obstruction of Bagchi’s criterion. The sharpest new formulation, isolated in the amplification analysis, is a distinction between two kinds of autonomy of a quadratic functional: its *value* can be computed from the primes alone, but its *inertia* (the partition of its negative index into blocks) cannot be determined without the positions of the zeros. We state the master quantifier as a synthesis principle, exhibit each closure as an instance, and present the unconditional ledger the programme leaves behind—among them the strict single-scale trace criterion, the tensor index identity $\text{neg. ind}(Q_1 \otimes Q_2) = p_1 q_2 + q_1 p_2$, an unconditional sub-resolution positivity theorem, and a quadruple-density target weaker than RH. We close with an honest account of what a proof must now do: cross the mean-to-uniform barrier by a mechanism no current argument-form provides.

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1. INTRODUCTION

1.1. The end of a programme, and what it leaves. A long programme [5, 6, 7] set out to prove RH and did not. What it produced instead is a map: a sequence of attempts, each documented and each either refuted or reduced to a named wall, culminating in the structural no-go of [6]. That paper argued that RH always appears as the boundary value of a one-parameter family whose interior is an unconditional theorem, that no one-dimensional propagation principle reaches the boundary, and that exactly three argument-forms survive: two-dimensional propagation (Form A), index amplification (Form B), and an effective version of Bagchi's self-approximation criterion (Form C). The companion paper [7] pursued all three. Forms A and C closed there; Form B survived a first decisive experiment and was reduced to a single missing lemma, the sieve-to-index bridge. That bridge has since closed [5,].

This paper does two things. First (§2), it records the three closures as a single ledger of theorems. Second, and principally (§3–§4), it argues that the three are one wall in three coordinate systems, isolates that wall as a single quantifier inversion, and gives its sharpest formulation. The remaining sections present the unconditional results the programme leaves behind (§5) and an honest account of the boundary of current methods (§6).

1.2. What is claimed. We claim no proof of RH and no conditional result following from a weaker hypothesis than is already known. The central claim is structural and is stated as a synthesis principle (Principle 3.1), not a theorem: that the obstruction to RH, across every arc of the programme, is the inversion of one quantifier. Several *instances* of the principle are theorems (the closure theorems of §2); the principle itself is the assertion that they are the same theorem. The one genuinely new conceptual contribution is the value/inertia distinction of §4, which we believe is the most precise form yet of the Hadamard barrier.

1.3. Framework. We use the notation of [4, 5, 6, 7]: $d\nu \geq 0$ is the unconditional discrepancy measure with RH $\iff d\nu = 0$; the bridge theorem [3] gives $\kappa(Q) = \text{neg.ind}(H_C) = 2m$ on the off-critical subspace $\mathcal{K}_{\text{off}}$, with m the number of off-critical quadruples and RH $\iff m = 0$; $\mathcal{W}_2$ is the two-variable Weil functional of [5,].

2. THE TRICHOTOMY, CLOSED

We restate the three closures as theorems, citing their proofs. Each identifies the precise point at which the form fails.

Theorem 2.1 (Form C: the self-blocking amplifier [5,], [7, §2]). *The self-approximation density $\delta(d, \varepsilon)$ does not degenerate as $d \rightarrow 0$; the guaranteed scale degenerates linearly, $T_0 \asymp C(\varepsilon)/|d|$. The Rouché amplification that would convert one off-critical zero into a positive-density family is bounded, under $\neg\text{RH}$, by an anchored density carrying the same exponent $\kappa(\sigma)$ as the unconditional zero-density ceiling $N(\sigma, T) = O(T^{\kappa(\sigma)})$ that would contradict it. The density needed to force RH and the ceiling that caps it move together under any improvement; mobile $d = d(T)$ does not separate them.*

Theorem 2.2 (Form A: degeneration [5,], [7, §3]). *Let $\mathcal{T}(t, x)$ be the joint heat-time/arithmetic discrepancy trace. The only definition for which both edges—vanishing for $t \geq \Lambda$ and reality of zeros at finite x —are unconditional theorems vanishes identically on the interior of the quadrant, because the forward De Bruijn–Newman flow preserves reality of zeros [2]. Hence $\partial_t \mathcal{T} \equiv \partial_x \mathcal{T} \equiv 0$ and every coupling inequality is vacuous; the corner $(0, \infty) = \text{RH}$ remains a pure discontinuity.*

Theorem 2.3 (Form B: inertia is not autonomous [5,]). *The negative index amplifies exponentially and unconditionally, $\kappa(\psi^{\otimes k}) = \frac{1}{2}(4m)^k$ on $\mathcal{K}_{\text{off}}$ [5,], and the tensor square is realised by the two-variable Weil functional $\mathcal{W}_2$ on a thickened multiplicative diagonal, whose value is computable from the primes alone [5,]. Nevertheless the window indices of $\mathcal{W}_2$ are binary—0 on every window under*

RH , and $\asymp m \log T$ without bound under $\neg RH$ —so any uniform finite bound on the relative negative index already implies RH at $k = 2$, before amplification can act. The bound the bridge requires is an asymptotic evaluation of the almost-prime functional with relative error $o(1/\log T)$ uniform over all windows, which implies RH directly.

The three forms fail at three apparently different places: a matching of density exponents (C), an invariance of the heat flow (A), a binarity of window indices (B). The rest of the paper argues they are the same place.

3. THE MASTER QUANTIFIER

3.1. The common shape. In each closure, separate the object that is unconditionally controlled from the object equivalent to RH .

Form	Unconditionally controlled	Equivalent to RH
C	zero density <i>on average</i> , $N(\sigma, T) = O(T^\kappa)$	anchored density in the <i>individual</i> window J_d
A	the trace on the <i>interior</i> / forward flow	the trace at the <i>boundary</i> corner $(0, \infty)$
B	the functional value <i>in mean</i> over windows	the index <i>uniformly</i> over every window

In all three the controlled quantity is an average, a generic value, or an interior value; the RH -equivalent quantity is the corresponding individual, specific, or boundary refinement. The gap between the two columns is, in each row, the inversion of one quantifier.

Principle 3.1 (The master quantifier). Across the programme, RH is equivalent to a statement obtained from an unconditional theorem by inverting a single quantifier:

“for almost all / on average / in the interior” $\longrightarrow$ “for the specific / uniformly / at the boundary.”

Every unconditional method certifies the averaged side; none certifies the refined side, because the arithmetic object sits in the measure-zero exceptional set of its own generic behaviour, in the favourable direction, and genericity cannot detect a measure-zero event.

We do not claim Principle 3.1 as a theorem; it is the synthesis the closures support. Its force is that it is not specific to the trichotomy: it organises the entire programme.

3.2. The principle organises the whole programme. The same quantifier inversion is visible in every arc.

Arc A (the parity barrier). Sieve methods control the count of integers with given congruence data *on average over residue classes*; RH -strength statements need to *distinguish* $\omega(n) \bmod 2$ for the individual sequence. Selberg’s parity principle is exactly the statement that the averaged data cannot see the individual sign [5, Arc A].

Arc B (the arithmetic-propagation wall MW-2). The Euler product controls ζ *generically* in the half-plane $\operatorname{Re}(s) > 1$; the zeros are *individual* global objects in the strip, owned by no prime. The information does not propagate because it is averaged information about a region, not pointwise information about a locus.

The CCM Hadamard barrier MW-7. No property of ζ verifiable without the positions of the zeros—i.e. no *generic*, position-blind property—implies $dv = 0$, the *individual* vanishing.

Bagchi’s criterion. The Bohr-torus flow is uniquely ergodic, so the *generic* orbit recurs; RH is the recurrence of the *individual* arithmetic orbit ω_0 , which lies in the measure-zero exceptional set [5,].

Observation 3.2. The four obstructions MW-2, MW-7, the parity barrier, and the Bagchi individual-orbit gap, previously catalogued as distinct walls, are four instances of Principle 3.1. The trichotomy closures (Theorems 2.1–2.3) are three more. The programme’s seven walls collapse to one.

3.3. Why the inversion is hard in exactly this way. The inversion is not a technical gap to be narrowed by sharper estimates. An averaged statement and its individual refinement differ by the behaviour on a null set, and *the favourable direction of RH is precisely a null-set event*: that the off-critical count is not merely $o(\text{generic})$ but exactly zero. No method that proceeds by majorisation, equidistribution, or density—all of which see only the averaged side—can certify an exact-zero individual event. This is why every unconditional bound in the programme has “the wrong sign” (MW-4): it bounds the average, and the average is compatible with both RH and its negation. The quantifier, not the size of the constant, is the obstruction.

4. AUTONOMY OF VALUE VERSUS AUTONOMY OF INERTIA

The amplification analysis produced the sharpest formulation of the wall, and we isolate it because it is the paper’s one new conceptual tool.

Definition 4.1. Let $\mathcal{W}_2$ be the two-variable Weil functional and Q_2 its associated quadratic form on the thickened-diagonal test space. We say the *value* of Q_2 is autonomous if $Q_2(h, h)$ is computable from arithmetic data (primes, pole, gamma factor) without the positions of the zeros of ζ . We say the *inertia* of Q_2 is autonomous if the partition of its negative index into the relative block $\text{neg.ind}(Q|_{\mathcal{K}_{\text{off}}^{\otimes 2}})$ and the remainder is so computable.

Theorem 4.2 (Value autonomous, inertia not [5, −108]). *The value of Q_2 is autonomous: the mixed zeros $\times$ primes terms reabsorb into arithmetic $\times$ arithmetic terms, and the primes $\times$ primes block is the almost-prime functional $\sum_N g(N) \sum_{d|N} \Lambda(d) \Lambda(N/d) \varphi(d^2/N)$, computable from the primes alone. The inertia of Q_2 is not autonomous: the negative directions live, at second order, in the deficit between the almost-prime kernel and its archimedean prediction—a pair-correlation quantity—and the isolation of the off-critical block from the off $\times$ on contamination requires knowing which zeros are off the line. The global negative index is ∞ under $\neg\text{RH}$; only the relative block is finite, and the relative block is not test-accessible.*

Remark 4.3 (The Hadamard barrier, sharpened). Theorem 4.2 is the Hadamard barrier MW-7 in its most precise form. Earlier statements asserted that no position-blind property implies $d\nu = 0$. The value/inertia distinction locates the obstruction one level finer: one *can* write down a position-blind functional whose value encodes the discrepancy (the barrier does not forbid that), but one *cannot* read its index—the very invariant that detects m —without the partition into blocks, and the partition is position-dependent. The quantity is autonomous; the *signature* is not. This is why the amplification route, which needs the index and not the value, meets the wall even though the value is in hand.

Observation 4.4. Value/inertia is the master quantifier in operator-theoretic coordinates: the value is the averaged (trace-like) datum; the inertia is the individual (spectral-resolution) datum. Computing a trace is averaging; resolving a signature is individuating. The same null-set obstruction reappears: the off-critical contribution to the value is $o(1/\log T)$ per window—generic zero—while its contribution to the inertia is a full negative direction—an individual event the average cannot see.

5. THE UNCONDITIONAL LEDGER

A programme that does not reach its goal is judged by what it proves along the way. We list the unconditional results, with sources, that survive independently of RH.

- (L1) *Non-negativity of the discrepancy*: $d\nu \geq 0$ as a measure, each off-critical quadruple contributing a factor ≥ 1 [5,].
- (L2) *Single-scale trace criterion*: the Abel kernel W_λ is strictly positive at every real point, whence $\text{RH} \iff T_\lambda[\lambda_0] = 0$ for a single λ_0 ; the universal quantifier of the criterion is redundant [5,], [6, Thm 3.2].

- (L3) *Tensor index identity*: for indefinite quadratic forms of signatures (p_i, q_i) , $\text{neg. ind}(Q_1 \otimes Q_2) = p_1 q_2 + q_1 p_2$, strictly superadditive; hence exponential amplification $\kappa(\psi^{\otimes k}) = \frac{1}{2}(4m)^k$ on $\mathcal{K}_{\text{off}}$ [5,], [7, Thm 4.2–4.3].
- (L4) *Realisation of the tensor square*: the two-variable Weil functional on the thickened diagonal realises $(\mathcal{K}_{\text{off}}, Q|_{\mathcal{K}_{\text{off}}})^{\otimes 2}$ with the predicted $(8, 8)$ signature, the thickening weight carrying the index [5,], [7, Thm 5.4].
- (L5) *Sub-resolution positivity*: $Q_2 > 0$ on test functions of fixed support in high windows; the negative index is invisible below the resolution scale $a \asymp \log T$ [5, Thm 3.3].
- (L6) *Closed-form kernel*: for $N(\lambda_0) = 1$, $B_{\lambda_0}(r) = \frac{1}{8} \text{sech}^{1/2}(r)(105 \text{sech}^4 r - 78 \text{sech}^2 r + 5)$; its prime series diverges like $(5\sqrt{2}/8) \log X$, so the CCM test function sits on the boundary of the Weil class [5,], [7, §3].
- (L7) *Eight equivalent reformulations* of RH as functionals of $d\nu$ [5,], and the bridge theorem $\kappa(Q) = \text{neg. ind}(H_C)$ [3].
- (L8) *A quadruple-density target*: a bound of the form $\kappa_W \leq C \cdot n_W$ on window indices would give $m \leq C/2$, an unconditional bound on the number of off-critical quadruples weaker than RH [5, §8.2]—behind the same wall, but a genuine partial target.

These are not consolation. (L2) and (L3) in particular are clean, quotable facts; (L5) is an unconditional positivity with a precise resolution threshold; (L8) is the shape of a theorem strictly weaker than RH that the framework could plausibly reach.

6. THE BOUNDARY OF CURRENT ARGUMENT-FORMS

6.1. What the closures establish jointly. [6] predicted that if Forms A and B also collapsed, the collapse would itself be informative: it would be the first evidence that RH is unreachable by any of the argument-forms currently known to mathematics. We have reached that point, and the evidence is now specific rather than impressionistic. Every form—propagation (1D in [6], 2D in Form A), amplification (Form B), recurrence (Form C and Bagchi), positivity (MW-1, the whole of Arc B)—terminates at the same quantifier inversion. The walls are not many; the wall is one, and we can now write it down.

Principle 6.1 (The boundary). A proof of RH must certify an individual, uniform, or boundary statement that its averaged, generic, or interior shadow leaves undetermined—an exact-zero event on a null set of the relevant generic behaviour. No argument-form catalogued in this programme (propagation, amplification, recurrence, positivity, sieve) does this, because each proceeds through the averaged shadow. The missing idea must cross the mean-to-uniform barrier by a mechanism outside this list.

6.2. Where such a mechanism might live. We will not pretend to know the mechanism. We can say what property it must have, sharpened by Theorem 4.2: it must read the *inertia* of an arithmetic functional, not merely its value—it must individuate a signature where only a trace is unconditionally available. In the one historical case where RH-type statements were proved (function fields), the mechanism that did this was cohomological: finite-dimensional ℓ -adic cohomology supplies an integer signature directly, bypassing the averaged analytic data entirely. The programme’s repeated arrival at the arithmetic site, the scaling-site square, and the Riemann–Roch analogue [5, MW-5] is, in this light, not a coincidence but a recognition: a genuinely cohomological invariant over $\text{Spec } \mathbb{Z}$ would supply the inertia non-generically, and that—not any analytic refinement—is what the master quantifier demands. Whether such an invariant exists is the Connes–Consani programme’s open question, and it is outside the analytic methods this programme exhausted.

7. ASSESSMENT

The programme set out to prove RH and did not; this paper records why, as precisely as we are able. The structural trichotomy of [6] is closed in all three branches (Theorems 2.1–2.3), and the three closures, with the programme’s earlier walls, are one obstruction: the inversion of a single quantifier from averaged to individual (Principle 3.1), in its sharpest form the distinction between the autonomous value and the non-autonomous inertia of a quadratic functional (Theorem 4.2). The programme leaves an unconditional ledger (§5) and a clear statement of what a proof must do (Principle 6.1).

We state the honest conclusion plainly. We have not narrowed the distance to a proof of RH; the distance is, if anything, more visibly large than when we began. What we have narrowed is the *description* of the obstruction: from a list of seven walls to a single quantifier inversion with a named operator-theoretic form. A reader who hoped for progress toward the theorem will find none here. A reader who wants to know exactly where the difficulty lives, and why every analytic, spectral, and combinatorial route returns to the same place, will find that this is now answerable in one sentence—and that the answer points outside the methods we used, toward an invariant that reads inertia rather than value. Mapping the wall is not crossing it. But a wall whose single gate is known is a better thing to leave behind than a fog.

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